ALGORITHMS AND DEBUGGING

How to solve a problem systematically without a computer.

SCAN & LEARN

WHAT HAPPENS • FOLLOW THE ARROWS

01 GET A TASK → 02 CHOOSE CLEAR STEPS → 03 FOLLOW THE STEPS → 04 CHECK THE RESULT

FOUR IDEAS TO KEEP

1 ALGORITHM = PRECISE RECIPE

A finite set of clear steps for a problem. Input is what comes in; output is what comes out. Instructions must be unambiguous.

Example: given 3 numbers → find the largest.

2 DECISIONS AND LOOPS

Decision: “if raining, take an umbrella”. Loop: “for each book, read its title”. Both are building blocks of programs.

“If...” = condition. “For each...” = loop.

3 USE A TRACE TABLE

When code changes values, write the current state after each step. Do not guess by glancing at the code.

step 0: x=1 | step 1: x=2 | step 2: x=4.

4 THREE KINDS OF BUG

Syntax: invalid instruction. Runtime: the program crashes. Logic: it runs but gives a wrong answer.

Debug: reproduce → find the first wrong step → check.

TRY IT — HUMAN ROBOT

Tell a classmate how to draw a square without pointing. Which instructions were missing?

THE ONE SENTENCE TO REMEMBER

A program is not “magic”: you can SIMULATE it one step at a time.

Question: Wrong answer, no crash = what kind of error?